

Soumyajyoti Dutta

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LLM Post-Training | Alignment | Evaluation Design | Agentic | RAG Systems | Applied ML for Cyberdefense

Education

Texas A M University

Ph.D. in Computer Science, Department of CSCE

College Station TX

Spring 2024 – Present

- **Advisor:** Dr. Marcus Botacin (Botacin's Lab). **Thesis:** LLMs for automated rule generation and model interpretability in cyberdefense.

Texas A M University

M.S. in Computer Engineering, Department of ECE

College Station TX

Fall 2022 – Fall 2023

- **Coursework:** ML Theory, Deep Learning, Data Mining, Randomized Algorithms, ML-Based Cyberdefenses.

SRM University

B.Tech. in Electronics and Communication Engineering

Chennai India

Fall 2017 – Spring 2021

Expertise

Core Competencies:

- **LLM post-training pipelines:** curriculum/staged SFT, structural reinforcement, reward-signal design, and behavioral shaping for fine-tuning and alignment.
- **Evaluation frameworks:** custom reward metrics (lexical/syntactic/semantic), LLM-as-judge and user-simulation, human-curated benchmarking across **10+ LLM families** plus cross-modal vision-language eval.
- **Distributed & efficient training:** **140M–9B-param** families (incl. sparse-MoE) on multi-node SLURM clusters; DeepSpeed / FSDP / DDP-NCCL / CUDA; rank-stabilized LoRA; vLLM with multi-turn KV reuse.
- **Agentic & RAG systems:** multi-tool dispatcher architectures, layered query routing, confidence-threshold anti-hallucination with web-search fallback, MCP-based tool integration.
- **Large-scale data pipelines:** multi-source dataset construction (**100M+ labeled examples**), tokenizer design, checkpoint management, and reproducible experiment tracking.
- **Systems & full-stack engineering:** CUDA profiling, memory-efficient loaders, FastAPI/React apps, local LLM orchestration, and ATS/scrapper pipelines.

Technical Stack:

ML: PyTorch, TensorFlow, HuggingFace Transformers, TRL, scikit-learn, LightGBM/XGBoost, NumPy, Pandas, Matplotlib

Distributed/Infra: DeepSpeed, FSDP, DDP/NCCL, CUDA, SLURM, vLLM, AWS **Serving/Agentic:** Ollama, MCP servers,

RAG (FTS5 retrieval), Serper/Semantic Scholar tool backends **Languages:** Python, C++, Rust **Web/Tooling:** FastAPI,

React/Vite/Tailwind, Django/Flask, SQLite, Git, CI/CD, Playwright **Domain:** YARA, IOC/NER extraction, malware analysis

(PE features)

Professional Experience

Texas A&M University — Botacin's Lab

Research Assistant / Machine Learning Engineer

College Station, TX

Spring 2024 – Present

- Built curriculum-based post-training pipelines across **140M–9B-param** families (BART, T5, Gemma 3, LLaMA 3.x, Nemotron; incl. sparse-MoE) on **multi-node SLURM clusters** with DeepSpeed, FSDP, and CUDA.
- **YaraGen (Booz Allen Hamilton-sponsored)** — end-to-end LLM system for automated YARA threat-detection rule generation (*USENIX Security 2026, in prep*):

Built a **20-tool agent dispatcher** with a **4-layer query router** and a RAG pipeline with confidence-threshold anti-hallucination, web-search fallback, and a Nemotron-4B synthesizer.

Constructed a **three-source dataset framework (100M+ labeled examples)**: hand-written synthetic curriculum, grammar-parsed procedural rules, and **305K** crawled real-world rules.

Designed **YaraAST**, a hybrid AST-aware tokenizer (**10 token reduction**); post-trained models to **≥99% syntax / ≥95% logical** validity on 22,655 held-out rules — a floor held even by BART-base 140M.

Built **Scooper** (BiLSTM hex extractor) and **MultiRuleGen** (IR-level merger) for rules over the 1,024-token window, raising corpus coverage **83%→100%**.

Designed **Brownie & Pu**, a reproducible reward/eval function (**BLEU + parsing + SAT semantics**) used as both training signal and quality gate; benchmarked **10+ LLM families** over **150+ runs**.

Built a custom **NER pipeline** for IOC extraction from unstructured analyst reports, reaching **92.5% F1** across **25 IOC types**.

- **AutoPYara** — open-sourced rule-generation package (PyPI) (*ACSAC 2026, submission*):

Re-implemented biclustering-based rule generation from raw binaries (**71K Windows PE samples**); benchmarked clustering algorithms with optimal-K heuristics.

Beat the **AutoYara SOTA** by **+14% coverage**, **+10% threat-hunting accuracy**, and **+8% low-sample generalization**.

- **Client-embedded consultant** for TJX Companies (Fortune 500); coordinated with client engineering/product stakeholders to deliver a **logistics-automation platform**.
- Optimized **REST API endpoints** for latency and concurrency; maintained **CI/CD** and led code review across cross-functional teams.

RESEARCH ENGINEERING PROJECTS

Ambiguity-Aware / Proactive Research Agent | *Co-authored — led model development training* 2026

- Added an explicit “ask” action to a deep-research agent, teaching it to **pause on underspeci ed queries** and ask one clarifying question before answering.
- Trained **Gemma-4-E4B** (sparse MoE, ~9B/4B) on a **single 80GB GPU** (vLLM, KV reuse) on the Dr. Tulu sca old with **MCP-exposed** web + Semantic Scholar tools.
- Designed a **structured XML tag protocol** and an ambiguity taxonomy (language vs. task) to make the pause decision **supervisable**.
- Ran a **four-stage SFT curriculum** (rank-stabilized LoRA): baseline → reflex pause (AmbigQA) → reasoned pause → mechanism-augmented distillation from **Claude Opus 4.6**.
- Built a **176-query benchmark** with three-axis eval; lifted **pause-detection F1 0.168→0.573** and diagnosed the remaining bottleneck as answer generation, not the decision to ask.

Sturdy Fishstick — Self-Built Job & PhD Search Platform | *FastAPI, React/Vite/Tailwind, Ollama* 2025

- Built a self-hosted, **dual-mode** job-search dashboard scoring listings with **fully on-device local LLMs** (qwen3:1.7b, LFM2.5 via Ollama) — no external inference API.
- Integrated **multi-source ingestion**: Google Jobs, LinkedIn, GitHub repos, **ATS APIs** (Greenhouse, Lever, Ashby, +3), Playwright fallback, and academic boards.
- Designed a **categorical-question scoring scheme** mapping deterministically to a 0–10 score — far more reliable for small local models than direct numeric scoring.
- Built an **FTS5-grounded RAG chat** that answers from exact DB counts (no hallucinated stats); added time-boxed scoring with model eviction, a Kanban tracker, and remote access.

Multimodal & Vision-Language Evaluation | *ECEN 642 — Digital Image Processing Computer Vision* 2024

- Evaluated a **DeepSeek vision-language model** against **human-annotated ground truth**; built the comparison harness and error analysis — a **cross-modal eval** exercise.

Machine Learning Malware Detection Model | *1st Place, CSCE 689 ML-Based Cyberdefenses Competition* 2022

- Built an end-to-end **Windows PE classi er** (EMBER/BODMAS/Benign-NET, **2M+ samples**) with LightGBM/XGBoost — **96.2% accuracy, 0.97 F1** under 1GB/5s constraints.
- Designed the competition’s **best-performing adversarial evasion suite** against peer defense models.

Multiperspective Hawkeye | *Systems / Computer Architecture*

- Implemented the **ISCA’16 Hawkeye** cache-replacement policy in **zsm**; extended with PC-tracking and multi-policy benchmarking.

HelloPentagon | *Explainable ML Malware-Defense Chatbot*

- Built an interactive analyst tool combining an **ML malware classi er** with **LLM-based triage** and interpretability.

Other Projects:

WildFire (spatiotemporal ML for wildfire-spread prediction from climate + satellite data),

Carotid Artery USG Analysis (CNN ultrasound classification — *Springer book chapter, 2022*),

Fashion-MNIST (CNN classifier comparing ResNet/VGG with regularization analysis).

PUBLICATIONS

- **YaraGen** — USENIX Security 2026 (submission in preparation).
- **AutoPYara** — ACSAC 2026 (submitted); open-sourced on PyPI.
- **Training and Benchmarking Pro-active Deep Research Agents to Overcome Underspeci city in User Queries** — manuscript in preparation, 2026 (co-authored, 4-person team).
- Carotid-artery ultrasound CNN classification — Springer book chapter, 2022.

TEACHING OUTREACH

Teaching Assistant, CSCE 439/704 — Data Analytics for Cybersecurity

Texas A&M University

Clustering, supervised ML, anomaly detection, security-focused visualization

Spring 2026

- Designed assignments and mentored student projects; supported lectures and grading.

Partner & Presenter, TAMU CS Day

Texas A&M University

Interactive AI/cybersecurity demos for high-school students

2024, 2025

- Designed and led hands-on demos introducing AI and cybersecurity concepts to non-specialist audiences.

Teaching Assistant, ECEN 325 — Electronics

Texas A&M University

Lectures, grading, and lab sessions

Spring 2023